

Exam History of AI

Date: Thursday May 22, 2025	Lecturer: dr. D.J. Beckers
Time: 2h 15min	Course code: FEW XB 0113

In general:

- (1) Make sure your name and student number are on ALL your work.
- (2) Answer in English sentences. Spelling and grammar should be correct either by English or by US standards. Just a few buzzwords never constitute an answer. Always explain yourself. Use appropriate examples to illustrate your answer.
- (3) This exam consists of 8 A-questions, 12 B-questions, 6 C-questions and a NO BONUS question. The A-questions are about the lectures; the B-questions refer to information from the book by Campbell-Kelly. The C-questions require you to combine the information from the book and lectures in your answer.
- (4) You pick four questions: 1 A-question, 2 B-questions and 1 C-question. Each question is worth 2 points. Indicate clearly which questions you're answering by mentioning either the number and / or the title of the question in your answer. If you answer more A-questions, only the first will be considered. Likewise for the B- and C-questions.
- (5) If you're *not* entitled to the bonus point, you additionally do the NO BONUS question. This question is worth one point.

A-Questions:

1. Technology and trust

During the lecture, (general) trust in technology was connected to the late 19th, early 20th century. Explain the line of thought, and describe how this trust was established.

2. Dutch dinosaurs

Discuss a history of one of the "Dutch dinosaurs". Make a comparison to one of the US or UK "dinosaurs".

3. Metaphor

Elaborate on the rise of the 1950s metaphor of "electronic brain" to refer to computers. To what extent was that metaphor related to research in AI?

4. AI in mathematics

In what ways was AI instrumental in 1960s/70s mathematics? Elaborate on two specific examples!

5. AI and education

In the 1960s and 70s AI was made instrumental in education. Elaborate on two specific examples.

6. Game culture

Discuss in what ways pc game culture and AI research interacted with each other in a mutual beneficial way.

7. History of AI

In the lectures, history of AI was connected to the rise of the field of AI – comparisons were made with history of computing. Elaborate on two of those comparisons.

8. Web 2.0 and AI

To what extent did the rise of Web 2.0 influence the rise of AI?

B-Questions:

1. Maurice Wilkes

What roles does Campbell-Kelly claim for Maurice Wilkes in his history of computing?

2. Computer manufacturers

Campbell-Kelly describes how, in the early 1950s, the automatic computer was reconstructed to become a business machine. Describe how this happened.

3. IBM

Campbell-Kelly discusses the rise of IBM in the computer industry with the phrase “Watson was rational but wrong, while Eckert and Mauchly were irrational but right”. Explain that phrase in the context.

4. Real time

Campbell-Kelly discusses the rise of real time computing as a military experiment that almost didn't happen and was not offering the safety it was supposed to create. Explain his line of thought.

5. Software crisis

To what extent was the late 1960s software crisis a real crisis? Discuss at least two stakeholders in the situation and explain their point of view.

6. Timesharing

Campbell-Kelly discusses the role of timesharing as essential for the rise of computing as something interesting for “ordinary people”. Explain his line of thought.

7. Personal computers

Campbell-Kelly suggest that personal computers became of interest because of a successful blend of two cultures, most outspokenly present in California. Explain his narrative. Which cultures does he refer to?

8. Broadening the appeal

According to Campbell-Kelly, the personal computer industry in the 1980s “broadened the appeal” of computers in several ways. Explain that remark. In your essay, mention the main firms relevant to the points you discuss.

9. Internet

Campbell-Kelly discusses the rise of Internet as the “confluence of three desires”. Which desires does he refer to, and how do these together explain the rise?

10. Diversity and inclusion

Campbell-Kelly addresses several instances of the computer industry in the 2010s starting to work in favour of diversity and inclusion. He mentions it as one of the more positive results (next to negative ones!) of the globalizing computer industry. Explain his line of thought.

11. Google

To what extent was Google offering something new with its search engine around 2000? In what sense does Campbell-Kelly relate Google to the rise of surveillance capitalism?

12. Computing and governance

In his final chapter, Campbell-Kelly addresses various ways in which 21st century computing became an ethical liability, which had to be (and sometimes was) mitigated by governments. Explain two of these in detail.

C-Questions:

1. Histories

The traditional history of AI, starting with Alan Turing in 1950 or with the Dartmouth conference in 1956 could be contested. Mention two alternative starting points and explain how these have merits as well. Use the concept of agendas to structure your essay.

2. Digital culture

To what extent is “digital culture” a suitable way of describing twenty-first century culture?

3. Silicon valley

In his chapter on the rise of Silicon valley, Campbell-Kelly decomposes the 1970s myth of “two boys and a garage” success stories. Still, in the lecture it was explained that it is hard to find successful examples of non US companies at the time. Discuss the two lines of development (Silicon valley and outside the US), comparing a successful valley and a non-valley company, addressing the concept of company culture.

4. Internet

The Internet boom of the early 21st century was discussed from different points of view in the book and the lectures. Elaborate on one of the main differences, using the concept of appropriation.

5. Defining technology

Discuss two 1960s examples of cybernetics as a defining technology (in Bolter's sense).

6. AI winters

Compare the narrative of the AI winters to the narrative of Babbage, that Aiken used to obtain funding at Harvard for building the Mark I computer. In your essay, use the concept of agendas.

NO BONUS

Elaborate on the roles of education in a history of AI between 1990 and 2010.